SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1704

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I MITHRA DEVI S - 192524203 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S. Mithra Devi

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared. **(b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked. **(b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings? **(c)** Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus. **(b)** If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightag e (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understan din g of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understan din g with minor gaps	Partial understandin g; misses key ideas	Lacks understan din g of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganize d and difficult to understand

Task 1: Article Summary

(a) Full Citation, Domain and ML Problem

Full Citation:

Moslehi, S., Rabiei, N., Soltanian, A.R. and Mamani, M. (2022). *Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran*. **BMC Medical Informatics and Decision Making, 22**, 192. **DOI:** 10.1186/s12911-022-01939-x.

Domain/Application Area:
Healthcare and medical decision support.

Specific ML Problem:
The study focuses on using machine learning to **classify COVID-19 patients according to mortality outcome** and identify the most important demographic, clinical, and laboratory factors associated with death.

The authors compared several decision-tree-based methods:

- Logistic Model Tree (LMT)
- C4.5
- C5.0
- CART

Random Forest was also used for **feature-importance analysis and feature selection**.

(b) Key Objective and Research Gap

The main objective of the article is to develop and evaluate interpretable machine-learning models for predicting mortality among COVID-19 patients. The researchers aimed not only to classify patients as recovered or deceased but also to identify the clinical and laboratory factors that have the greatest influence on mortality. The study addresses a research gap in earlier COVID-19 prediction studies, where many machine-learning approaches concentrated mainly on prediction accuracy while providing limited interpretability of the factors responsible for the prediction. To address this gap, the authors analyzed patient information collected from hospitals in Hamadan, Iran, and initially considered 25 demographic, clinical, and laboratory variables. Random Forest was used to determine feature importance, after which six important variables were selected. These variables were then used with LMT, C4.5, C5.0, and CART models. The models were evaluated using measures such as accuracy, recall, and F1-score, while ROC-AUC was also considered. The authors found CART to be the most effective model among the tested approaches and demonstrated that decision-tree methods can provide useful and relatively interpretable support for COVID-19 mortality assessment.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

1. Random Forest

Random Forest was used mainly for **feature selection**. It consists of multiple decision trees and determines the importance of different input variables.

The authors initially considered **25 variables** and calculated their relative importance. Features with sufficient importance were selected for further classification.

The six important features identified were:

1. Gender
2. Age
3. Blood Urea Nitrogen (BUN)
4. Partial Thromboplastin Time (PTT)
5. Serum Glutamic-Oxaloacetic Transaminase (SGOT)

6. Erythrocyte Sedimentation Rate (ESR)

2. Logistic Model Tree (LMT)

LMT combines the concepts of **decision trees and logistic regression**. It creates a tree structure while using logistic regression models at the terminal nodes.

3. C4.5 Decision Tree

C4.5 is a decision-tree algorithm that selects attributes using **information gain ratio**. It can handle classification problems and includes pruning to reduce unnecessary tree branches.

4. C5.0 Decision Tree

C5.0 is an improved decision-tree algorithm designed for efficient classification. It supports pruning and provides efficient tree construction.

5. CART

CART stands for **Classification and Regression Tree**. For classification, it generally creates binary splits and uses impurity measures such as the **Gini index**.

CART produced the strongest overall performance in this study.

Dataset Description

Dataset Property	Description
Application	COVID-19 mortality prediction
Total patients	**2,470**
Initial variables	**25**
Selected variables	**6**
Data source	Sina and Besat hospitals, Hamadan, Iran

Dataset Property	Description
Period	February 2020 – July 2021
Target	Recovered / Deceased
Recovered	1,853
Deaths	617

The researchers examined demographic, clinical, and laboratory information. Data containing missing or invalid values were handled during preprocessing, and important features were subsequently selected using Random Forest.

(b) Mapping to CO4

CO4 Topic	Connection with Article
Inductive Learning	Models learn classification rules from previously observed COVID-19 patient data.
Decision Trees	C4.5, C5.0 and CART are directly applied for classification.
Statistical Learning	LMT incorporates logistic regression into the tree structure.
Reinforcement Learning	Not used in this study.

Methodological Strength

The major strength is **interpretability**. Decision trees can provide understandable classification rules, making the results easier for healthcare professionals to examine compared with many black-box models.

Methodological Limitation

A major limitation is the **limited geographical and institutional diversity of the dataset**. The data came from only two hospitals in Hamadan, Iran. Therefore, the model may not perform equally well on patients from different hospitals, countries, healthcare systems, or populations.

Task 3: Results & Critical Evaluation

(a) Key Results

The researchers compared four machine-learning models.

Model	Accuracy	F1-Score	Performance
LMT	76.89%	85.73%	Good
C4.5	77.70%	86.08%	Good
C5.0	76.89%	85.74%	Good
CART	**78.24%**	**86.81%**	**Best**

For the CART model, the reported **recall was 95.50%**, while the reported **AUC was 79.5%**.

Therefore, **CART was the best-performing model among the tested approaches**.

(b) Critical Evaluation

The reported results appear realistic for a medical classification problem. The accuracy of approximately 78% indicates that the model is useful but is certainly not perfect. The **95.5% recall** reported for CART is particularly important in a mortality-prediction context because identifying high-risk patients is more important than simply maximizing overall accuracy.

However, several issues should be considered.

Potential Biases

1. Dataset bias:

The data were obtained from only two hospitals in one geographical region. This can introduce population and hospital-specific bias.

2. Class imbalance:

There were considerably more recovered patients than deceased patients. Therefore, accuracy alone may not completely represent model performance.

3. Missing-data bias:

Removing records with missing or invalid values can potentially change the distribution of the dataset.

4. External validation:

The model would be more convincing if it were tested on an independent dataset from different hospitals.

Additional Experiments

The study could be strengthened by:

- Testing the model on an external hospital dataset.
- Performing k-fold cross-validation.
- Reporting confidence intervals for evaluation metrics.
- Comparing CART with Random Forest and Gradient Boosting.
- Performing calibration analysis.
- Testing the model on different demographic groups.
- Evaluating performance on more recent COVID-19 datasets.
- Performing explainability analysis using SHAP or similar techniques.

(c) Real-World Applicability

The proposed approach could be deployed in **healthcare decision-support systems**.

Possible Applications

- Hospital patient-risk assessment
- Emergency department triage
- COVID-19 mortality risk screening
- Intensive-care prioritization
- Clinical decision-support systems
- Hospital resource planning

For example, a hospital system could enter patient age, gender, BUN, PTT, SGOT and ESR values and use the trained model to estimate the patient's mortality risk.

Challenges in Industry Deployment

Data Availability:

Large, high-quality and standardized medical datasets are required.

Data Privacy:

Patient information must be protected according to healthcare privacy regulations.

Generalization:

A model trained in one hospital may not work equally well in another hospital.

Integration:

The model must be integrated with existing Electronic Health Record systems.

Ethical Issues:

Healthcare decisions should not rely completely on an automated model. Doctors should remain responsible for final decisions.

Model Drift:

Medical conditions, treatment procedures and patient populations can change over time, requiring periodic model evaluation and retraining.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

Insight 1: Inductive Learning

I learned how **inductive learning** uses previous patient examples to discover patterns and create a model that can classify new patients.

CO4 Connection:

This directly relates to the concept of **Inductive Learning**, where a general model is learned from specific training examples.

Insight 2: Decision Tree Classification

I understood that different decision-tree algorithms can produce different results even when they use the same dataset. C4.5, C5.0 and CART use different splitting and tree-building strategies.

CO4 Connection:

This connects directly to **Decision Trees**, including tree construction, attribute selection and classification.

Insight 3: Importance of Evaluation Metrics

I learned that accuracy alone cannot completely describe the performance of a classification model. Metrics such as **recall, F1-score and AUC-ROC** provide additional information.

CO4 Connection:

This connects to the evaluation of **statistical and machine-learning classification models**, especially when the classes are imbalanced.

(b) Proposed Novel Experiment / Enhancement

I would extend this research by developing an **Explainable Ensemble Learning Model for COVID-19 Mortality Prediction**.

Instead of using only individual decision trees, I would compare:

- CART
- Random Forest
- Gradient Boosting
- XGBoost
- Logistic Regression

The models would be evaluated on a **multi-hospital dataset** containing patients from different geographical regions.

The experiment would measure:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Calibration
- Confusion matrix

An explainability method such as **SHAP** could also be added to show which patient features contributed most to each prediction.

Feasibility

This experiment is feasible because the original study already identifies a small set of important clinical features. Using the same features makes the extended model easier to implement and compare.

Expected Impact

The proposed enhancement could improve **generalization, prediction reliability and interpretability**. Testing the model across multiple hospitals would also determine whether it can be safely used in real-world healthcare environments.